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A novel experimental approach to investigate the potential for behavioural change in sharks in the context of depredation

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ABSTRACT

Baited video systems have been widely used to assess the relative abundance and diversity of sharks in locations around the world, however they provide limited information on behaviour. We developed and pilot tested a novel experimental approach to investigate whether repeated deployments of baited video systems in the same location could generate quantitative data on shark behavioural patterns, in the context of shark depredation (where sharks consume hooked fish). Specifically, we sought to test whether repeated exposure to boats and food in the same location would lead to a change in the arrival time and first feeding time of sharks, over a short timescale. We used the Ningaloo Marine Park (NMP) in Western Australia, a location where higher shark depredation rates have been identified in consistently fished areas, as a case study. A modified Baited Remote Underwater Video (BRUV) system was repeatedly deployed at two fished sites and two sites within a no-take marine reserve in the NMP, over six consecutive days, to mimic repeated recreational fishing and the availability of hooked fish for sharks to depredate. This approach was designed to investigate and disentangle the potential role of changes in behaviour versus variation in shark abundance, as a mechanism for how and why shark depredation can occur. Here, we report preliminary results from this methodological approach, where time of arrival and time of first feeding declined markedly in the fished site over 6 days of BRUV deployments, compared to the control site in the no-take marine reserve. A greater number of individuals from four carcharhinid species were observed at the fished site, compared to only three individuals from two species in the no-take marine reserve. The preliminary results from pilot testing of this novel experimental approach suggest that, with further modifications to identify individual sharks, and a greater spatial and temporal replication of sampling, it may be possible to identify behavioural changes occurring in sharks in the context of shark depredation. Understanding this mechanism can bring important benefits for fishers and managers, as it can lead to modifications in fishing methods designed at reducing the occurrence of behavioural changes in sharks, and thus mitigating shark depredation.

1. Introduction

Baited video systems can provide novel insights into the abundance and diversity of sharks in their natural environment, using a simple and cost-effective methodology (Brooks et al., 2011; White et al., 2013; Harvey et al., 2018). However, the behavioural information generated by baited video deployments only represents a short snapshot of sharks' wider behavioural repertoire, limiting the potential for understanding patterns in behaviour. Deploying baited video systems many times at the same location in a consistent and repeated experimental design could offer an improved ability to detect behavioural patterns, as it can enable observation of the same individuals in the same environmental/habitat context over time. One application of this is to assess behavioural patterns where sharks are repeatedly provided with a food source in a predictable timeframe, over consecutive days. Shark depredation (where sharks consume hooked fish) is a scenario where

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repeated availability of food (hooked fish) in the same location may occur (Mitchell et al., 2018a), therefore repeated deployments of a baited video system could allow investigation of behavioural changes in the context of depredation.

Depredation of fishing catch by sharks occurs during demersal and pelagic fishing in both commercial and recreational fisheries, with estimated rates ranging from 0.9 to 26% (Gilman et al., 2007; Mitchell et al., 2018b). Depredation can cause injuries to sharks, loss of income for fishers and a reduction in the quality of the fishing experience (Mitchell et al., 2018b; Ryan et al., 2019). In the Ningaloo Marine Park (NMP) in Western Australia, a recent survey of recreational fishers reported that, in some areas, approximately 12% of hooked fish were depredated by sharks (Mitchell et al., 2018a). Additionally, higher densities of fishing vessels and greater cumulative fishing pressure in specific areas of the NMP, over time, were found to result in higher depredation rates, as well as the formation of depredation hotspots (Mitchell et al., 2018a). Higher depredation rates in discrete areas may reflect greater abundances of sharks due to high prey density or favourable habitat, or alternatively it could be attributed to behavioural changes in sharks. The formation of behavioural associations where sharks associate the regular presence of fishing vessels with the availability of hooked fish and offal to feed on, may be occurring in areas of high fishing pressure in this region (Mitchell et al., 2018a). The release of unwanted fish which are stressed from being exposed to air, suffering from barotrauma and have impaired reflexes (McArley and Herbert, 2014), may also represent an easy food source for sharks, contributing to such behavioural associations. Previous studies with captive sharks have documented changes in behaviour consistent with the formation of behavioural associations (Clark, 1959; Guttridge et al., 2009; Guttridge and Brown, 2014; Vila Pouca and Brown, 2018). Provisioning of wild sharks has also been shown to result in changes in behaviour, diet and movement patterns in some cases (Johnson and Kock, 2006; Bruce and Bradford, 2013; Brunnschweiler and Barnett, 2013; Brunnschweiler et al., 2018; Brena et al., 2015; Barnett et al., 2016; Gallagher et al., 2015), although not in others (Laroche et al., 2007; Hammerschlag et al., 2012; Abrantes et al., 2018), and recent research in New Caledonia suggests increased wariness and a lower likelihood of sharks biting bait in fished areas close to populations centres (Juhel et al., 2019). However, the potential for behavioural associations to occur in the context of shark depredation has not been investigated empirically, therefore research into this behavioural process is necessary to understand the mechanisms underlying depredation.

We sought to design and test a novel and non-invasive experimental approach to identify whether the repeated deployment of a baited video system in the same location can enable investigation of patterns in shark behaviour, in the context of the repeated provision of food from shark depredation. We hypothesised that by repeatedly deploying a modified Baited Remote Underwater Video (BRUV) system in a location that is regularly fished, we could identify potential changes in shark behaviour through the time it took sharks to arrive and feed on consecutive days. By comparing the abundance, arrival and first feeding times of sharks at this fished location versus a control site in a no-take marine reserve where no fishing occurs, we sought to further identify how repeated fishing in discrete areas can potentially change shark behaviour. We aimed to apply Generalised Additive Mixed Models (GAMMs) to investigate the influence of temporal variables and shark abundance on time of arrival and time of first feeding of sharks, thus enabling us to test the validity of this methodological approach for investigating behaviour in the context of shark depredation.

2. Materials and methods

For this preliminary study, we deployed a modified mid-water BRUV system (Fig. 1a,b) to record the time of arrival, time of first feeding and relative abundance of sharks in an area at the tip of North-West Cape, Western Australia (VLF Bay), which receives a medium level of fishing pressure (see Fig. 1a in Mitchell et al., 2018a), as well as within a ~ 7 km^2 no-take marine reserve (Bundegi sanctuary zone, established in 1987 (CALM and MPRA, 2005)), in the NMP (Fig. 2a,b). Two replicates sites were sampled within each of the fished area (21.799°S, 114.189°E and 21.785°S, 114.190°E) and the no-take marine reserve (21.858°S, 114.181°E and 21.872°S, 114.178°E), on six consecutive days between 12/04/2014 and 17/04/2014, with a distance of 1 km between sites to reduce the likelihood of sharks moving between them. The two sites in the fished area were approximately 1.3 km and 2 km from the coast, compared to 2.2 km and 2.7 km for the two sites in the no-take marine reserve. The depth was ~13 m for all sites, and the habitat type was predominantly low-lying reef and rubble. Each site was sampled three times each day at approximately 0900, 1200 and 1500, on each of the six sampling days, apart from day three (14/04/2014) when sampling at 1200 and 1500 was not possible due to bad weather. This resulted in a total of 16 deployments at each of the four sites.

The BRUV system consisted of two downward facing GoPro™ cameras mounted on a horizontal aluminium base bar approximately 0.7 m apart, inwardly converged at an angle of approximately 8 degrees (Harvey and Shortis, 1998). The base bar was affixed to a 1 m long vertical pole attached with rope at each end and held vertical using a suspended weight approximately 2.5 m below the cameras and surface buoy (Fig. 1). Two tuna heads and a plastic-coated wire mesh bait bag filled with 800 g of crushed pilchards (*Sardinops sp.*) were attached to a metal bar 2 m below the cameras (5 m below the surface). The tuna heads were easily accessible to mimic a hooked fish in a recreational fishing setting (Fig. 1). The tuna heads were consumed by sharks during five of the 12 BRUV deployments in the fished sites on day one of the sampling period, and then not at all during the other sampling days. After the heads were removed, sharks were still observed interacting with the bait bag, therefore the removal of the heads was not judged to have diminished the attractive potential of the BRUV setup to sharks. The vessel was anchored and the BRUV system was then attached to the stern of the vessel and left to record for 60 min at each site. We had no control of recreational fishing occurring in the fished areas during our study and recreational fishing vessels were observed between 100 and 500 m away on 10 occasions throughout the sampling period, except on day two. The greatest number of fishing vessels were observed on day six, with vessels sighted during five of the 12 deployments on this day. At one of these locations, four vessels were observed less than 500 m away. The vessels recorded on the other sampling days were all at least 500 m away.

All video footage was analysed in EventMeasure (SeaGIS 2016). Time of arrival was classified as the time a shark first entered the field of view and time of first feeding represented the time between the first arrival and the first bite of the tuna head by a shark. All sharks were identified to species level where possible and were verified by a second expert. The relative abundance of sharks (MaxN) was calculated as the maximum number of individuals of the same species seen in the field of view at one time (Priede et al., 1994; Cappo et al., 2004). Relative abundance was calculated for each species and for all species pooled. The influence of relative abundance (all shark species pooled) and temporal variables (Table 1) on time of arrival ($n = 30$) and time of first feeding ($n = 30$) of sharks (all species pooled due to low sample sizes) at the two fished sites, was tested using separate GAMMs (Lin and Zhang, 1999). Tidal height, current speed and sea surface temperature data for the sampling locations were also investigated, but were not included in the GAMMs, due to the low level of variation in these parameters over the short study period. The Tweedie distribution (Tweedie, 1984) was used for the GAMMs, due to the large number of low values which affected the distribution of the response variables. Site was included in the model as a random factor to account for variability between sites. To assess the importance of these predictor variables on time of arrival and time of first feeding, we used a full-subsets GAMM approach, which selects the combination of predictor

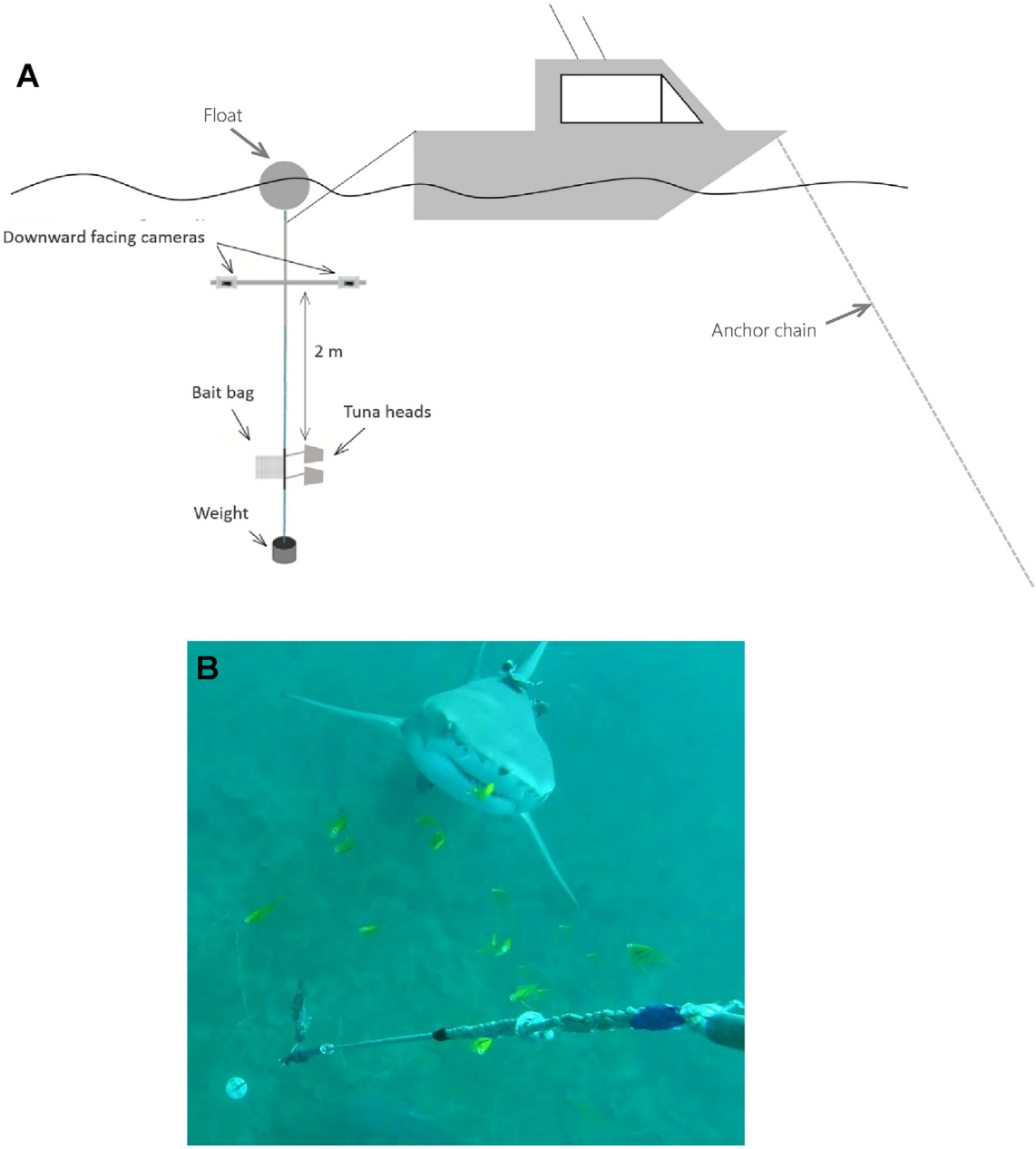

Fig. 1. A) Schematic of the modified Baited-Remote-Underwater-Video (BRUV) system deployed to record the time of arrival and time of first feeding of sharks, along with relative abundance (MaxN). Diagram not to scale. B) a sicklefin lemon shark *Negaprion acutidens* investigating the modified BRUVs system.

variables that produces the best-fitting, most parsimonious model (Fisher et al., 2018). We used Akaike Information Criteria (AIC) (Akaike, 1974) for model ranking and AIC weights to calculate importance scores of each predictor variable across all possible models (Burnham and Anderson, 2002). GAMMs were conducted using the 'FSSgam' package (Fisher et al., 2018). All analyses were conducted in the R language for statistical computing (R Development Core Team, 2015). Due to the low sample size for the two most common species (sicklefin lemon shark (*Negaprion acutidens*; Rüppell, 1837) and pigeye shark (*Carcharhinus amboinensis*; Müller and Henle, 1839)) we were unable to run separate GAMMs for these species.

3. Results

The experimental approach we utilised enabled successful collection of BRUVs footage from the fished and no-take marine reserve sites, over a consecutive 6 days period. Four shark species from two genera were recorded at the two sites in the fished area, all of which were from the family Carcharhinidae (Table 2). *Negaprion acutidens* was the most frequently observed species in the fished sites (Table 2). All shark species were observed feeding in the fished sites, although not all individuals of each species fed. Blacktip/Australian blacktip sharks (*Carcharhinus limbatus/tilstoni*; Müller and Henle, 1839/Whitley, 1950) had the fastest time of arrival (4.1 mins) and time of first feeding (5.6 mins), although this was based on a single individual (Table 2). *Negaprion acutidens* and *C. amboinensis* also had fast arrival and first feeding times (< 13 min), whereas tiger sharks (*Galeocerdo cuvier*; Péron & Lesueur, 1822) and dusky sharks (*Carcharhinus obscurus*; Lesueur 1818) had substantially longer arrival and first feeding times (> 35 min) (Table 2). In the no-take marine reserve, only three sharks were seen from two species, both of which were also from the family Carcharhinidae, and no feeding occurred (Table 2). Across both the fished site and the no-take marine reserve, it was not possible to distinguish individual sharks, due to lack of identifiable features such as scars,

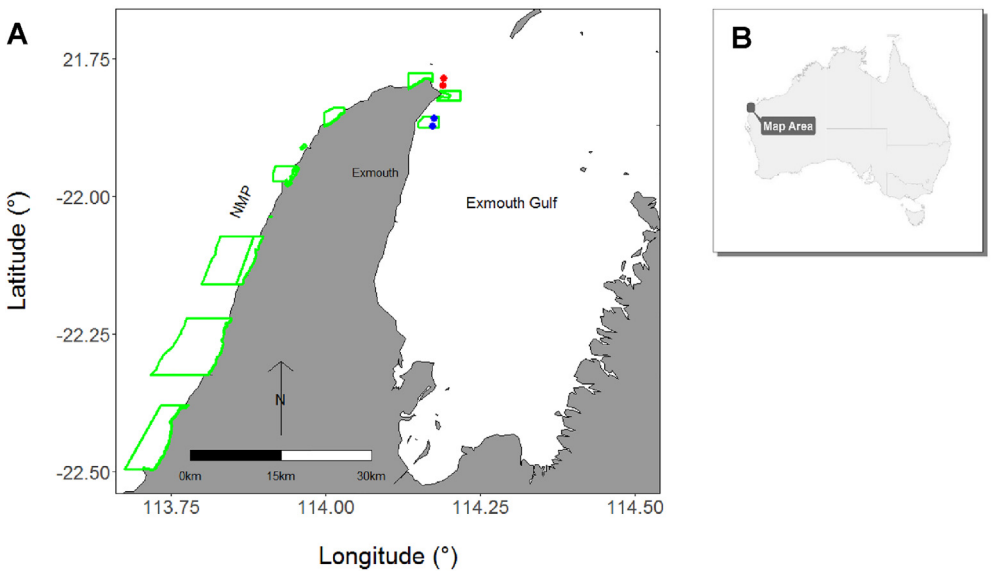

Fig. 2. A) Map of Baited Remote Underwater Video (BRUV) deployment locations in a fished area (red points) and a no-take marine reserve (blue points), in the Ningaloo Marine Park (NMP). Green lines indicate no-take marine reserve boundaries in the NMP. B) Location of the study site in north-west Australia. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

colouration patterns or other unique visual features.

There was a clear relationship of decreasing time of arrival and time of first feeding in the fished area, across the sampling period (Fig. 3a,b). Date/time was the only predictor variable in the best-fitting, most parsimonious models for time of arrival and time of first feeding and had a high relative importance in both models (0.99) (Fig. 4). Conversely, the relative abundance of sharks had a low importance in the models for time of arrival (0.19) and time of first feeding (0.17) (Fig. 4) at the fished sites. The time of day also had low importance (0.15 and 0.17, respectively, across both models) (Fig. 4). The low number of sharks recorded in the no-take marine reserve (Table 2) prevented analysis on either the time of arrival or the time of first feeding for these sites.

4. Discussion

The novel experimental approach we utilised in this study detected a rapid and marked change in the time of arrival and time of first feeding of sharks over consecutive days in the fished area, supporting our hypothesis that this methodological approach can yield important data on patterns of shark behaviour in the context of depredation. The theory behind using time of arrival as a behaviour metric is that a pattern of decreasing arrival time may reflect a short-term and localised change to behaviour consistent with associative learning, where the sharks associated the presence of the vessel with the availability of easily accessible food, arriving more quickly over consecutive days as the association is reinforced. Associative learning has been observed in elasmobranchs in captive studies where the provision of food was associated with a sensory cue (Clark, 1959; Guttridge et al., 2009; Guttridge and Brown, 2014; Vila Pouca and Brown, 2018). Furthermore, some studies have reported decreasing arrival times and changes in habitat use and site fidelity as a result of provisioning in wild white sharks (*Carcharodon carcharias*; Linnaeus, 1758) (Johnson and Kock, 2006; Bruce and Bradford, 2013; Gallagher et al., 2015; Brena et al., 2015) and whitetip reef sharks (*Triaenodon obesus*; Rüppell, 1837) (Fitzpatrick et al., 2011). The movement patterns of bull sharks (*Carcharhinus leucas*; Müller and Henle, 1839) also changed on a short-term basis at provisioning sites in Fiji (Brunnschweiler and Barnett, 2013). Mitchell et al. (2018a) recorded higher shark depredation rates in areas that were regularly and consistently fished, supporting the possible occurrence of behavioural change in sharks in the context of depredation. The chance of behavioural associations forming may be increased if areas of consistent fishing activity overlap with the small home ranges of certain species, such as *N. acutidens* (identified by Mitchell et al. (2019) to be responsible for depredation in the NMP), which can have a high fidelity to single reef systems (Filmalter et al., 2013).

The cue that could potentially enable associative learning to occur in the context of shark depredation may be sound, as this sensory cue propagates the furthest distance through water (Myrberg et al., 1972; Urick, 1979; Chapuis, 2017). The repeated occurrence of vessel engine noise in a specific location, in conjunction with a predictable food reward in the form of hooked fish, could therefore lead to the development of a behavioural association. By using the time of arrival and time of first feeding metrics, our approach tests this theory, by identifying whether sharks are responding to the noise and arriving more quickly as the behavioural association develops and becomes reinforced. Tracking data for grey nurse sharks (*Carcharias taurus*; Rafinesque, 1810) and *C. leucas* have shown that individuals may respond to the presence of fishing vessels by swimming towards them and spending extended periods circling them (McCord and Lamberth, 2009; Otway et al., 2009). Likewise, certain shark species have been found to display

Table 1
Predictor variables tested in the full-subsets Generalised Additive Mixed Models for the time of arrival and time of first feeding of sharks, and their hypothesised importance to the response variables.

Predictor variable	Hypothesised importance to the time of arrival and time of first feeding of sharks
Time since the start of sampling (date/time)	The repeated presentation of tuna heads in a simulation of recreational fishing activity, over 6 days and at the same sites, was expected to lead to sharks developing a behavioural association between the presence of the vessel and the availability of food, therefore resulting in consecutively faster arrival and first feeding times
Time of day	The movement patterns and feeding activity of sharks is known to vary across diel periods (Bush, 2003, Vianna et al., 2013), therefore this may have influenced their arrival and first feeding times.
Relative abundance (MaxN of all shark species pooled)	The relative abundance of sharks in the sites sampled may have influenced the time of arrival and time of first feeding recorded, because a higher number of sharks present would increase the likelihood of sharks being recorded quicker.

Table 2
Shark species identified in the fished area and the no-take marine reserve, their cumulative relative abundance (MaxN) across the whole sampling period, mean (+/− standard error; SE) time of arrival, whether they were observed feeding, and if so, the mean (+/− standard error; SE) time of first feeding.

Scientific name	Common name	Cumulative relative abundance	Mean (+/− SE) time of arrival (mins)	Observed feeding	Mean (+/− SE) time of first feeding (mins)
Fished sites					
Negaprion acutidens	Sicklefin lemon shark	33	8.2 (2.1)	Yes	12.4 (2.9)
Carcharhinus amboinensis	Pigeye shark	24	8.4 (3.4)	Yes	5.7 (2.4)
Carcharhinus obscurus	Dusky shark	2	39.6 (8.1)	Yes	47.7 (n/a)
Carcharhinus limbatus/tilstoni	Blacktip/Australian blacktip shark	1	4.1 (n/a)	Yes	5.6 (n/a)
TOTAL		60			
No-take marine reserve					
Galeocerdo cuvier	Tiger shark	2	50.2 (3.6)	No	n/a
Carcharhinus limbatus/tilstoni	Blacktip/Australian blacktip shark	1	1.2 (n/a)	No	n/a
TOTAL		3			

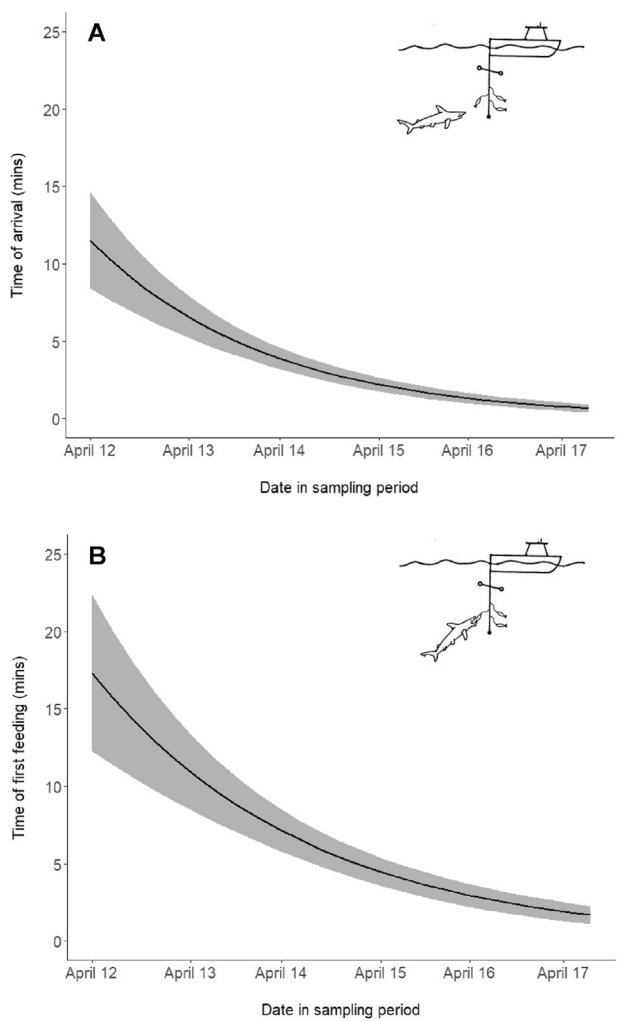

Fig. 3. The influence of time since the start of sampling (date in sampling period) on A) the time of arrival and B) time of first feeding of sharks as predicted by the most parsimonious Generalised Additive Mixed Models (GAMMs) for these metrics. Solid line = model predicted values, grey areas = predicted values ± 95% confidence intervals (2 standard errors). Illustrations in top-right of each plot indicate arrival and first feeding.

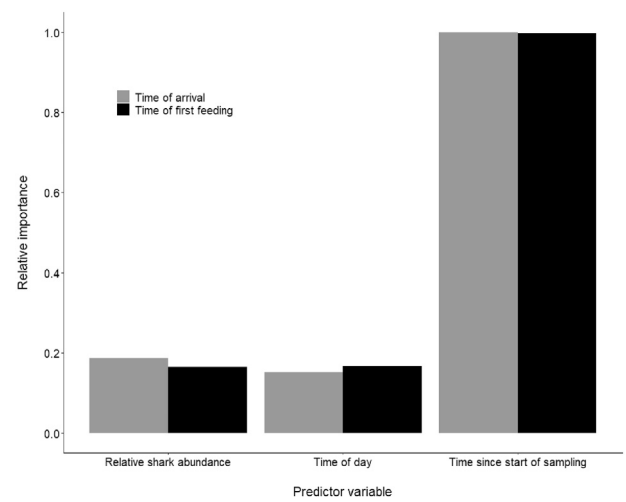

Fig. 4. Relative importance values for the predictor variables tested in the Generalised Additive Mixed Models (GAMMs) for time of arrival (grey bars) and time of first feeding (black bars).

directed movements to, and/or higher residency in, locations where provisioning regularly occurs (Brena et al., 2015; Fitzpatrick et al., 2011; Brunnschweiler and Barnett, 2013; Bruce and Bradford, 2013). The cue in these cases may therefore have been the sound generated by the vessel's engines and/or the echosounder or anchor chain.

Odour cues may also be an important stimulus for the development of these behavioural associations, however the arrival times of < 1 min we recorded at the end of the sampling period suggest that the sharks were already close to the vessel before the bait and tuna heads even entered the water, suggesting that auditory cues may have been used by the sharks to locate the vessel. The visual cue of a fish struggling on a hook, as well as the vibrations it creates, may also be important sensory mechanisms used by sharks to locate and depredate hooked fish during recreational fishing. The presence of multiple recreational fishing vessels < 500 m away during some of the BRUV deployments in the current study, particularly on day six of the sampling period, may have contributed to the fastest arrival times being recorded on this day, due to the concentration of sensory cues created by the vessels within a small area. This is supported by Mitchell et al. (2018a), who reported higher depredation rates with increasing numbers of vessels fishing in close proximity. Yet, the presence of other vessels was not considered to be a substantially important factor driving the marked decrease in time of arrival and time of first feeding of sharks across the sampling period overall, as this relationship was also observed across the days where little or no vessels were sighted. Future larger-scale studies that aim to investigate the behavioural response of sharks to repeated fishing in the same location should quantify the potential impact of nearby vessels, to further investigate how this may impact shark behaviour.

To more definitively identify if behavioural associations are occurring during the experimental approach we applied, it is necessary to

identify whether the same individual sharks are returning to the BRUV system on consecutive days, otherwise we cannot rule out whether it is different individual sharks passing through the area and opportunistically approaching the BRUV system on the different sampling days. Unfortunately, identifying specific individuals was not possible in the current study, due to the lack of distinguishing features on the sharks observed. However, future research could aim to tag sharks with externally visible colour-coded tags, to provide a feature that could be easily identified from BRUV footage. Another important factor to address is potential variation in abundance, to be able to disentangle the effects of behaviour and abundance. The low level of spatial and temporal replication in the current study prevented us from definitively assessing the role of shark abundance in the results achieved, especially considering the very low number of sharks observed in the no-take marine reserve. For example, despite our best efforts to deploy the modified BRUV in locations with similar habitat at the sites open and closed to fishing, it is possible that the habitat quality and/or abundance of natural prey species was higher in the fished sites, resulting in the greater relative shark abundance recorded. The low relative abundance we reported in the no-take marine reserve contrasts with other studies, which have found markedly higher shark relative abundance in reserves (Juhel et al., 2019; Frisch and Rizzari, 2019; Bond et al., 2012; Goetze and Fullwood, 2013; Speed et al., 2018). However, the relatively small size of the reserve in our study (~7 km^2), compared to those considered in previous research, which had areas > 50 km^2 (Bond et al., 2012; Goetze and Fullwood, 2013; Speed et al., 2018), may have influenced this result due to the lower level of overlap with shark home ranges. The low numbers of individuals of each species recorded at the fished sites in the current study, which required the pooling of species for GAMM analysis, reduced the ability of our study to detect differences in behaviour between species. Inter-specific differences may play an important role in determining the behavioural response of sharks to the BRUV system over time, due to variation in the boldness of different species (Clua et al., 2010), as well as the occurrence of inter-specific competition and the formation of hierarchies when multiple species are competing to access a food resource (Ritter, 2001; O'Shea et al., 2015; Papastamatiou et al., 2018). Environmental variables such as water temperature and tidal range have been shown to affect the movements and feeding behaviour of sharks (Carey et al., 1990; West and Stevens, 2001; Sims et al., 2006; Lowry et al., 2007; Speed et al., 2012) and current speed can affect the bait plume dispersal and its potential to attract sharks when BRUVs are used (Heagney et al., 2007). However, these variables were found to have little variation across the study period, so were not included in the analysis.

In light of the drawbacks that we identified in the current research, we propose that a future study could be conducted in the NMP, where there exists a strong gradient in recreational fishing pressure due to variation in access and the presence of an extensive network of no-take areas (Mitchell et al., 2018a; Smallwood and Beckley, 2012). These existing gradients of fishing pressure could be used to investigate and disentangle how shark behaviour and abundance are influencing rates of shark depredation on a broader spatial scale, and across seasons. The replication of our study methodology with a substantially increased level of sampling effort would enable GAMMs to be run for each individual species, thus providing the ability to detect potential differences in behaviour between and within shark species. Environmental variables such as temperature and current speed should also be included in the analyses, to account for their potential influences on shark activity levels, feeding behaviour and bait plume dispersal. The addition of a method for identifying individual sharks, such as through colour-coded external tags, would confirm whether it is the same sharks returning to the sampling location on consecutive days, adding weight to the hypothesis that the predictable and repeated availability of food leads to changes in shark behaviour. Also, identifying the extent of overlap between fishing vessel activity and shark movement patterns, using Vessel Monitoring System (VMS) and acoustic/satellite telemetry data, respectively (Queiroz et al., 2016, 2019), could be one approach to further investigate potential changes in shark behaviour and movement patterns due to the availability of fish to depredate. Overall, this study has identified and tested a novel approach for investigating potential behavioural changes in sharks that may occur due to the availability of hooked fish in specific locations. The results produced highlight the potential of our experimental method, and by conducting a large-scale study based on this approach, researchers could add further weight to the hypothesised occurrence of behavioural changes in the context of depredation, which have been widely reported anecdotally. By understanding the behaviour of sharks, fishers may be able to modify their fishing methods to reduce the likelihood of creating and reinforcing potential behavioural associations, thus reducing the occurrence of depredation.

Compliance with ethical standards

This research was conducted in accordance with Animal Ethics Approval RA/3/100/1317 from
The University of Western Australia.

Declaration of competing interest

The authors have no conflicts of interest to report.

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Credit author statement

J.D. Mitchell – Writing - original draft; Writing - review & editing; Investigation; Data curation; Formal analysis; Validation; Visualization.

M. Schifiliti – Funding acquisition; Investigation; Methodology; Validation.

M. Birt – Data curation; Funding acquisition; Investigation; Methodology; Project administration; Resources; Writing - review & editing.

T. Bond – Investigation; Methodology; Project administration; Resources; Writing - review & editing.

D.L. McLean – Conceptualization; Methodology; Supervision; Writing - review & editing.

P.B. Barnes – Conceptualization; Funding acquisition; Resources; Writing - review & editing.

T.J. Langlois – Conceptualization; Funding acquisition; Investigation; Methodology; Project administration; Supervision; Writing - review & editing.

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